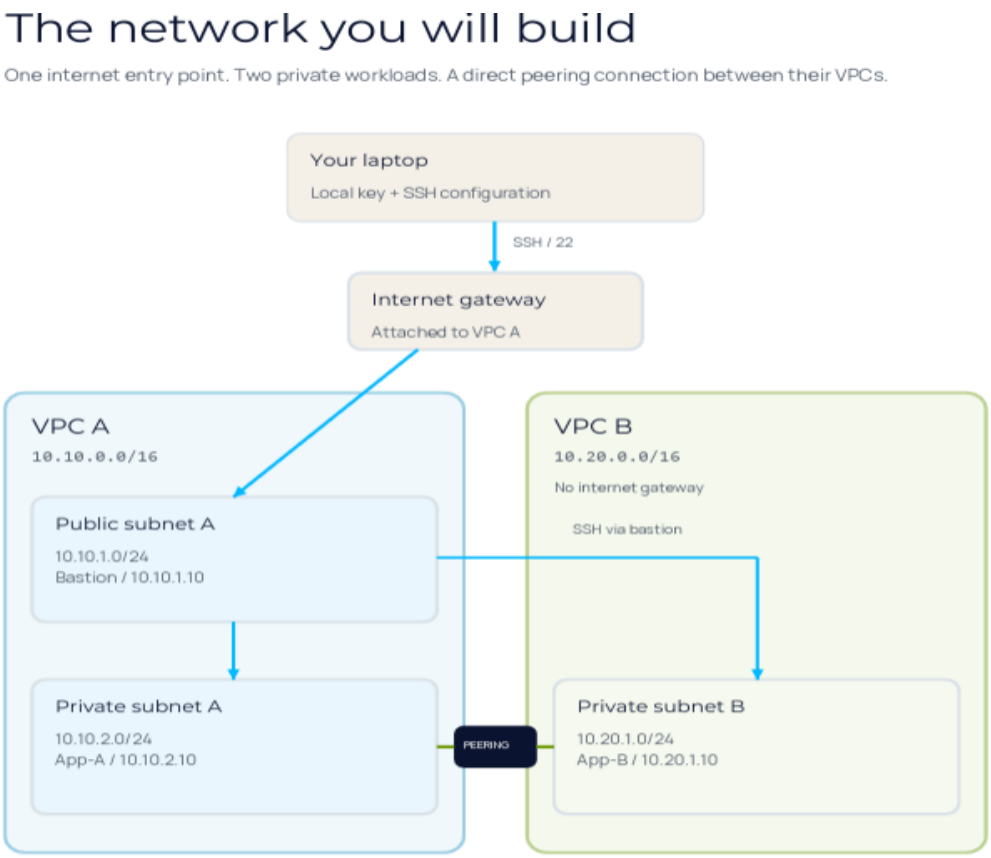
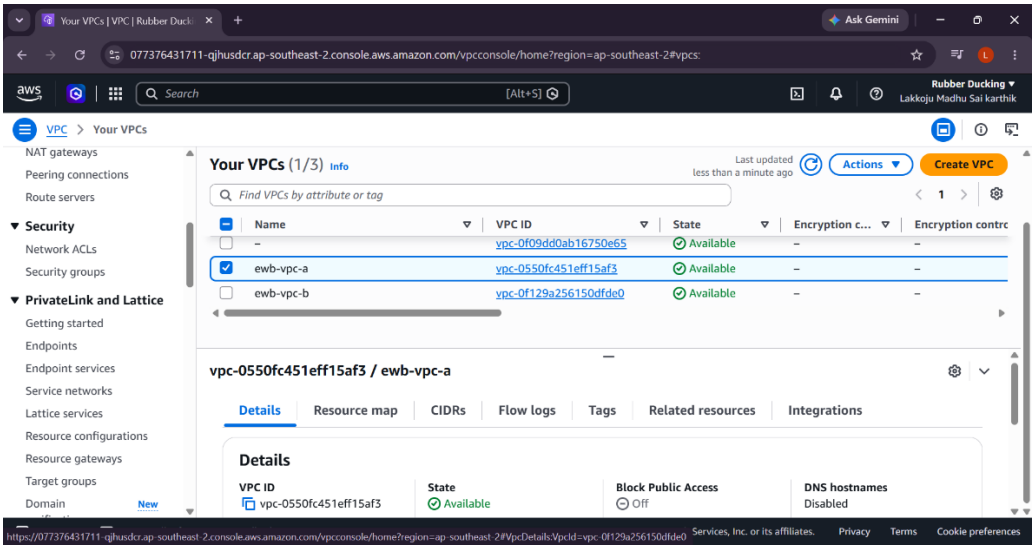
- **Project Title: EWB VPC Peering Lab**
- **Student Name: KAVALA NAGA VEERA MANIKNATA**
- **Batch: Batch 1**
- **Region: ap-southeast-2(Sydney)**
- **Date : 09-11-26**

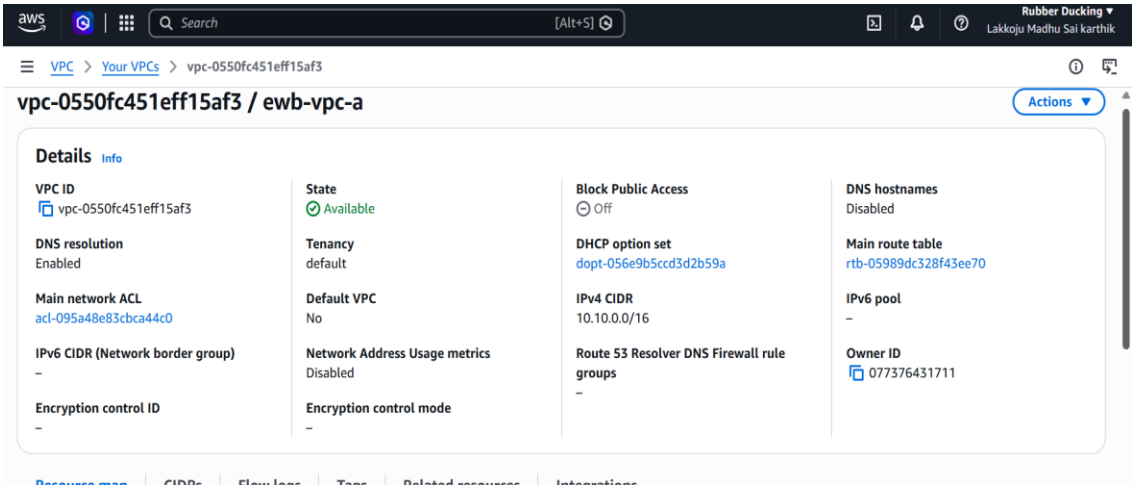


Resource	Name / ID	CIDR / IP.
VPC A	ewb-vpc-a	10.10.0.0/16
VPC B	ewb-vpc-b	10.20.0.0/16
	Bastion Instance ID	10.10.1.10
	App-A Instance ID	10.10.2.10
	App-B Instance ID	10.20.1.10
	Peering ID	pcx-03d72875a0fec9fc1

	Security Group IDs	**sg-03c4ffbaf19013a2a - ewb-sg-bastion**



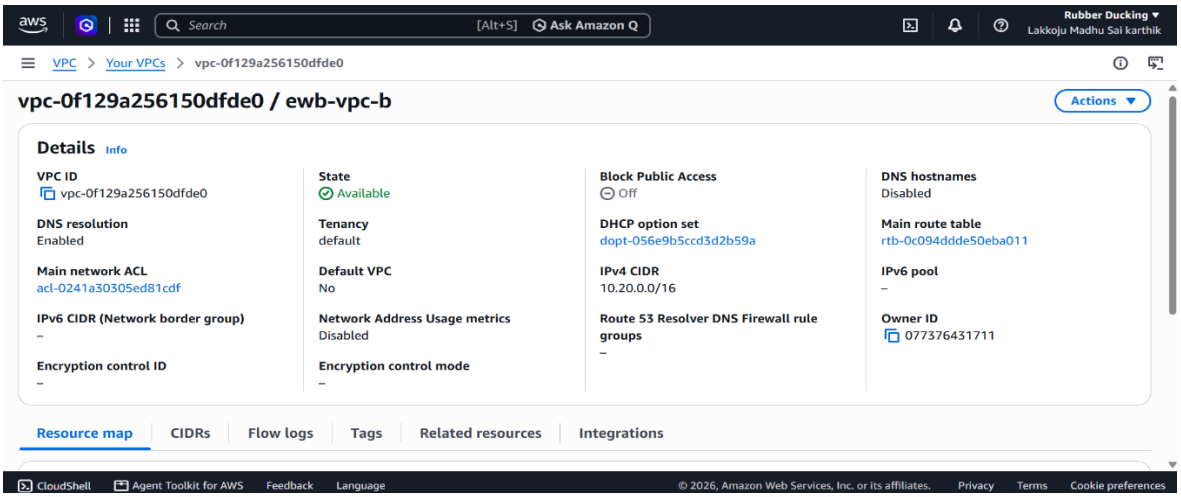
THE VPC'S WE CREATED



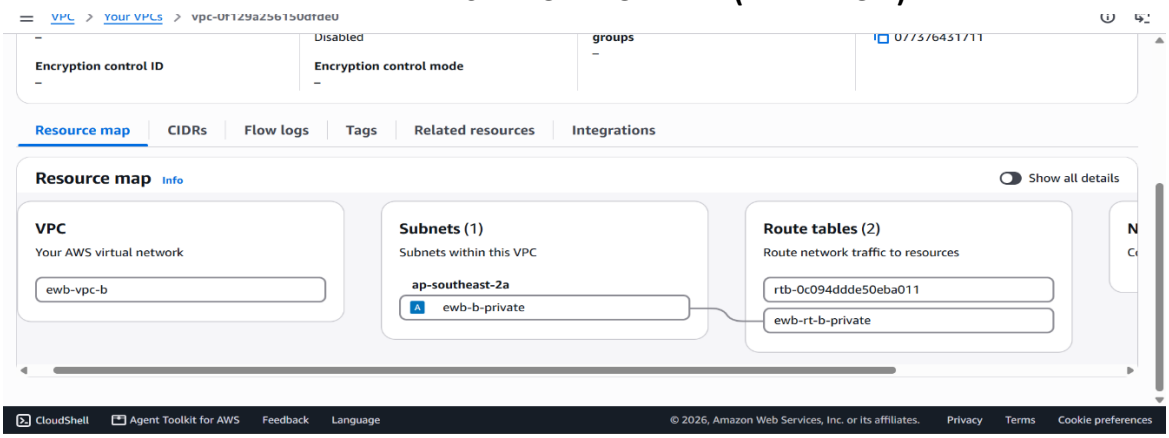
THE FIRST VPC WE CREATED(EWB-VPC-A)



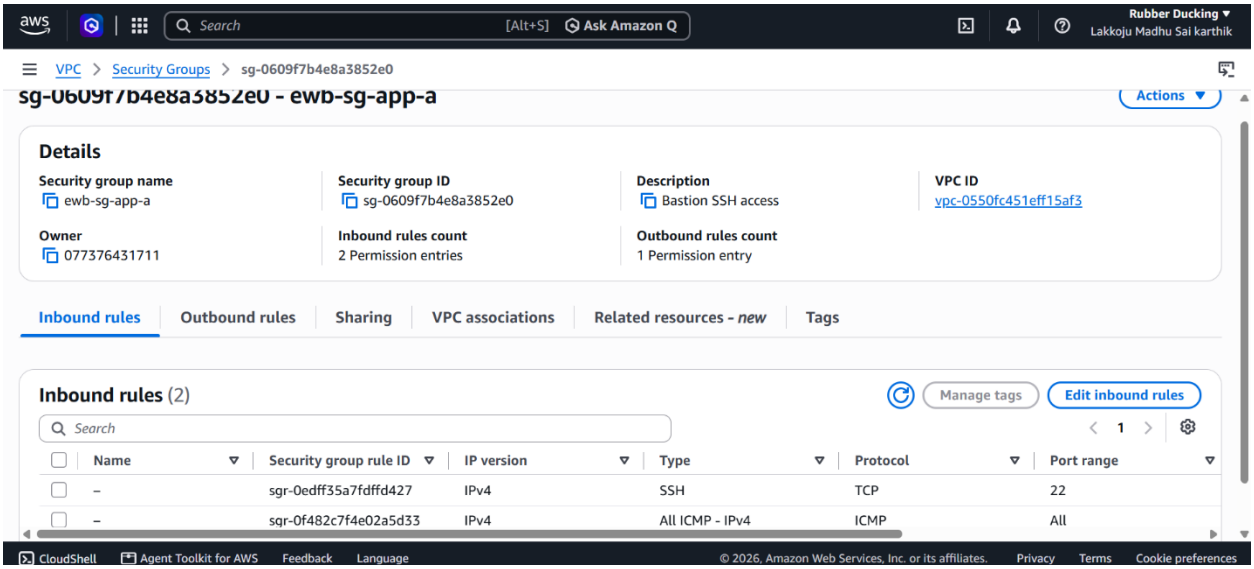
THE SUBNETS AND ROUTE TABLES OF EWB-VPC-A



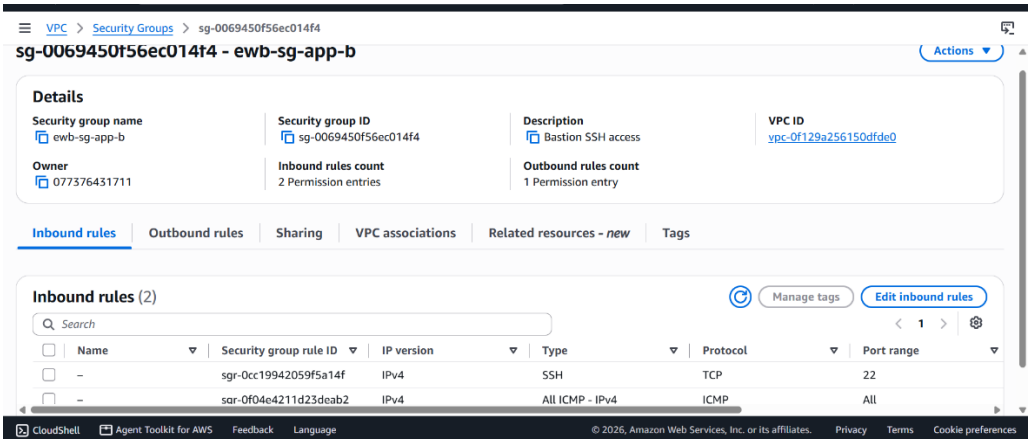
THE FIRST VPC WE CREATED(EWB-VPC-B)



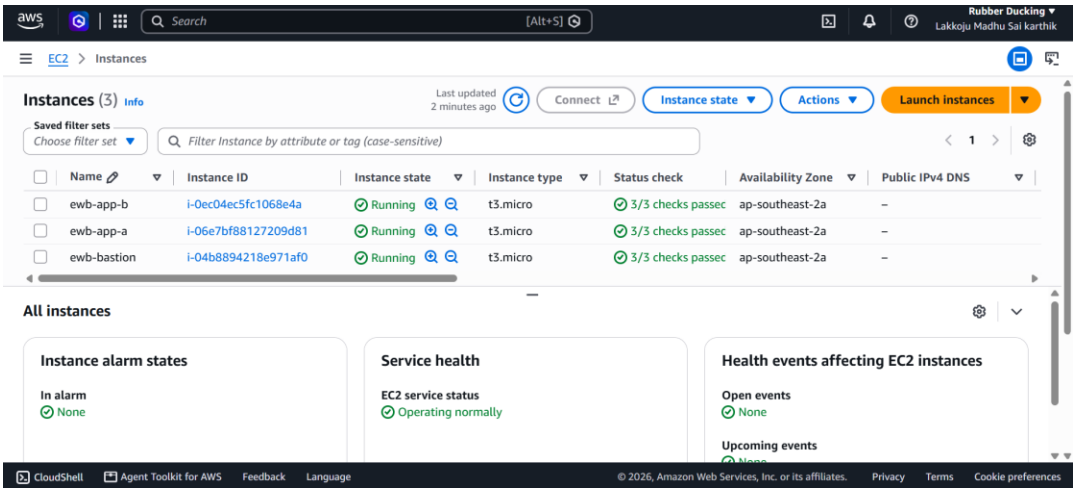
THE SUBNETS AND ROUTE TABLES OF EWB-VPC-B



SECURITY GROUPS



S **SECURITY GROUPS**



EC2 INSTANCES WE CRTEATED

```
[ec2-user@ip-10-10-1-10 ~]$ ssh -i ~/ewb-lab.pem 10.20.1.10
The authenticity of host '10.20.1.10 (10.20.1.10)' can't be established.
ED25519 key fingerprint is SHA256:HpnUuvVGWPPCTzz8ikeV9fTAWrMjh6QLXDVSbBS786o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.153 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.186 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.180 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.168 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3155ms
rtt min/avg/max/mdev = 0.153/0.171/0.186/0.012 ms
[ec2-user@ip-10-20-1-10 ~]$ exit
logout
Connection to 10.20.1.10 closed.
[ec2-user@ip-10-10-1-10 ~]$ ssh -i ~/ewb-lab.pem 10.10.2.10
```

0% PACKET LOSS

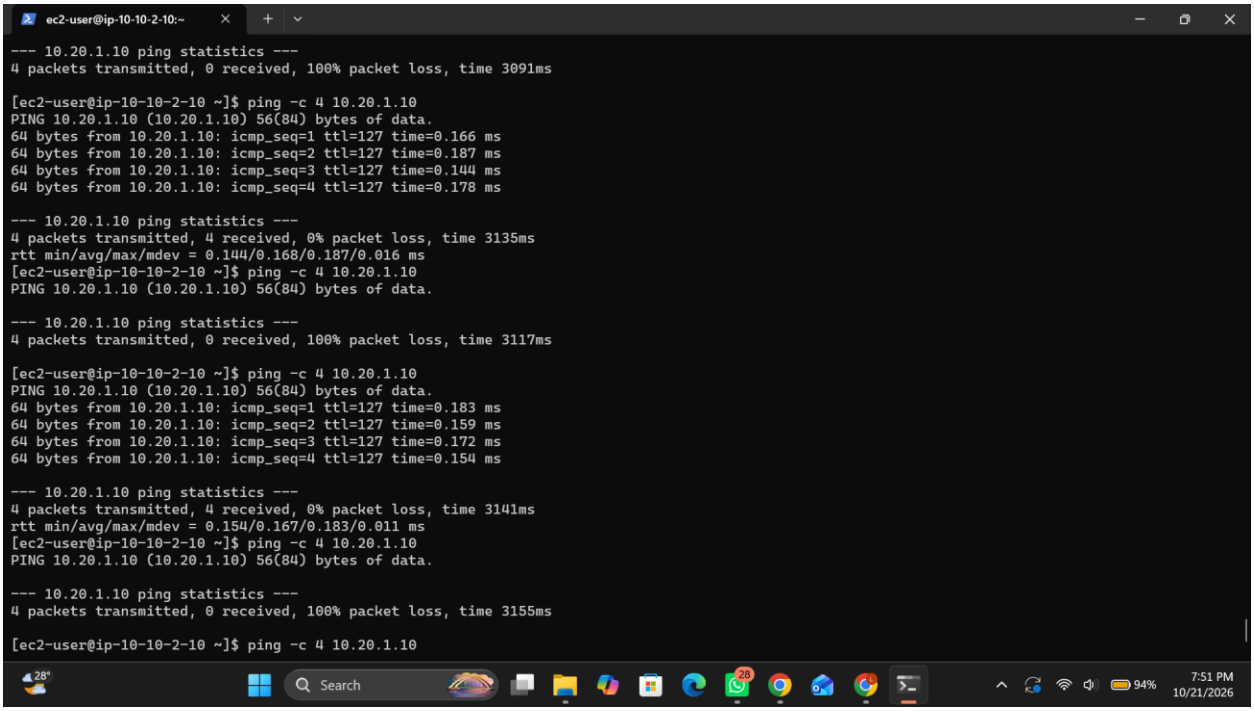
```
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3155ms
rtt min/avg/max/mdev = 0.153/0.171/0.186/0.012 ms
[ec2-user@ip-10-20-1-10 ~]$ exit
logout
Connection to 10.20.1.10 closed.
[ec2-user@ip-10-10-1-10 ~]$ ssh -i ~/ewb-lab.pem 10.10.2.10
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Wed Oct 21 13:38:50 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3147ms

[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.170 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.158 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.213 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.182 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3152ms
rtt min/avg/max/mdev = 0.158/0.180/0.213/0.020 ms
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
```

Controlled Failure (100% packet loss after deleting the rule)



Baseline Failure (100% packet loss before peering)

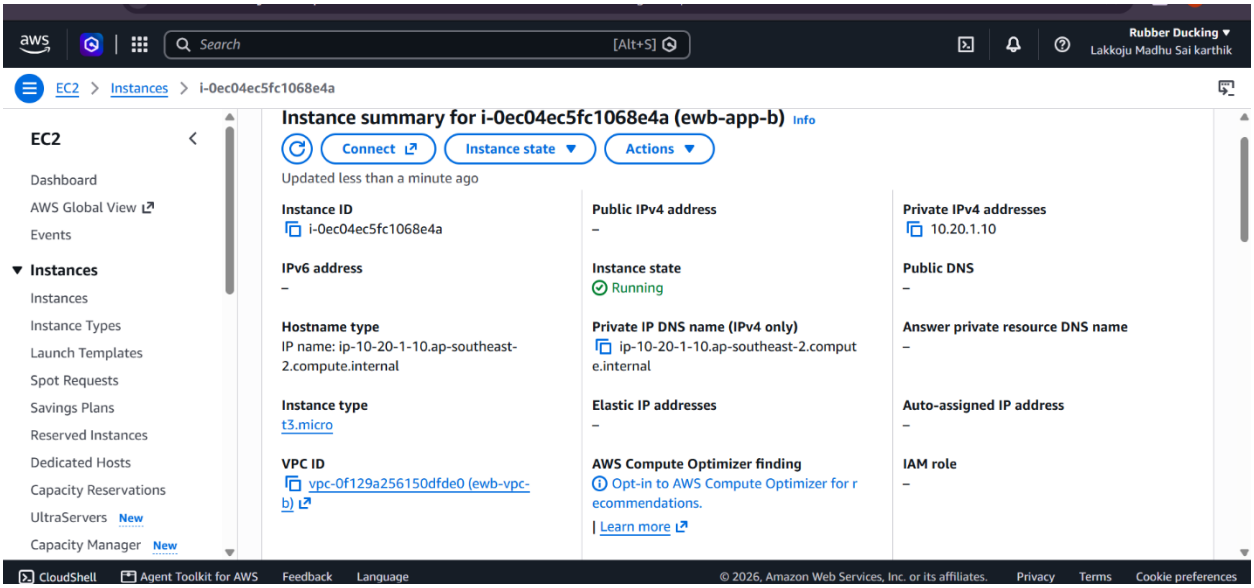
```
4 packets transmitted, 4 received, 0% packet loss, time 3141ms
rtt min/avg/max/mdev = 0.154/0.167/0.183/0.011 ms
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3155ms

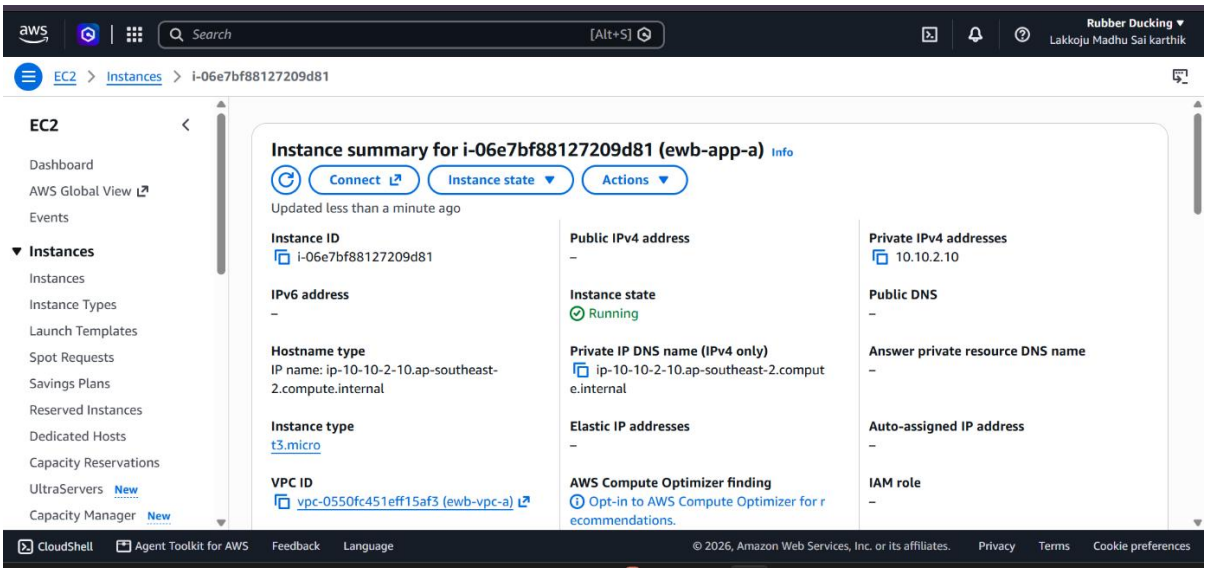
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.275 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.144 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.199 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.159 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3131ms
rtt min/avg/max/mdev = 0.144/0.194/0.275/0.050 ms
[ec2-user@ip-10-10-2-10 ~]$
```

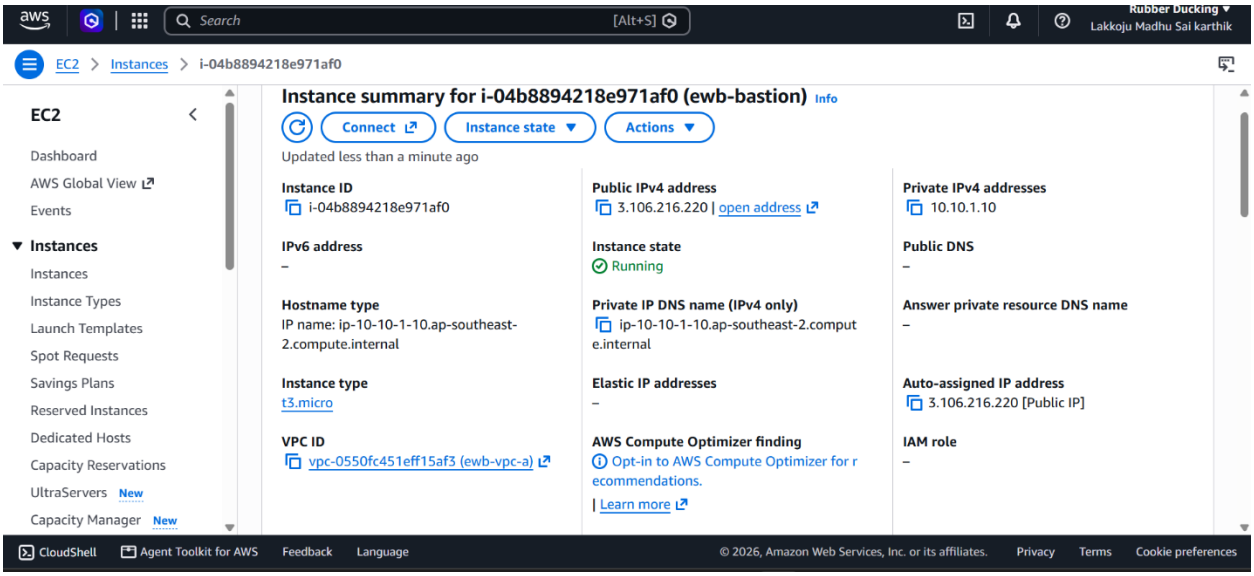
Recovery (Ping works after adding the rule back)



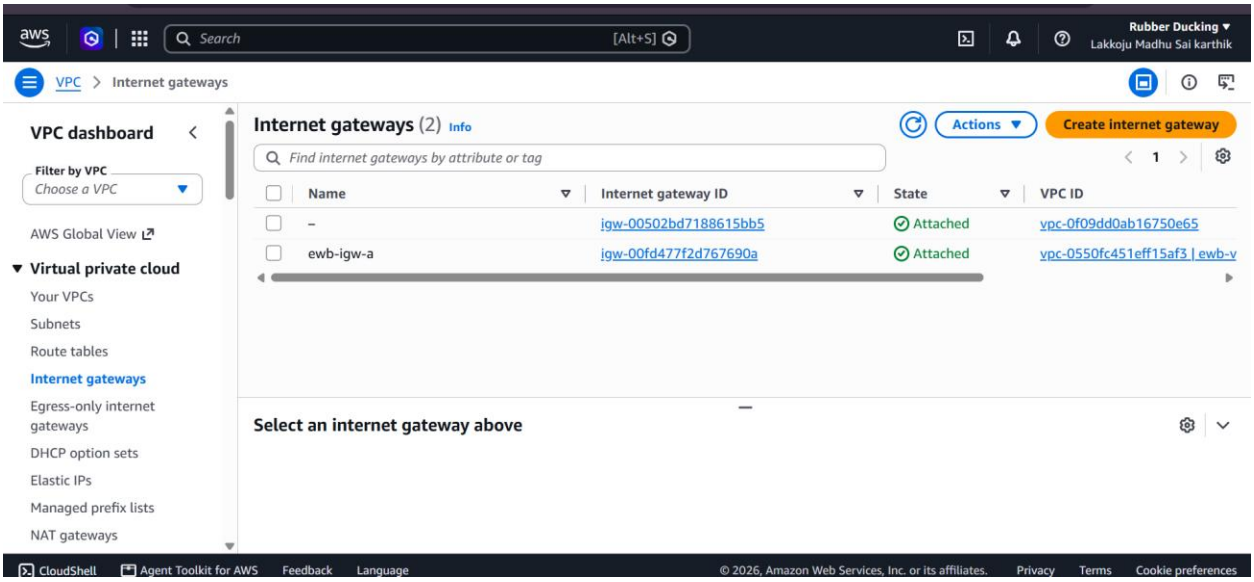
INSTANCE OF EWB-APP-B



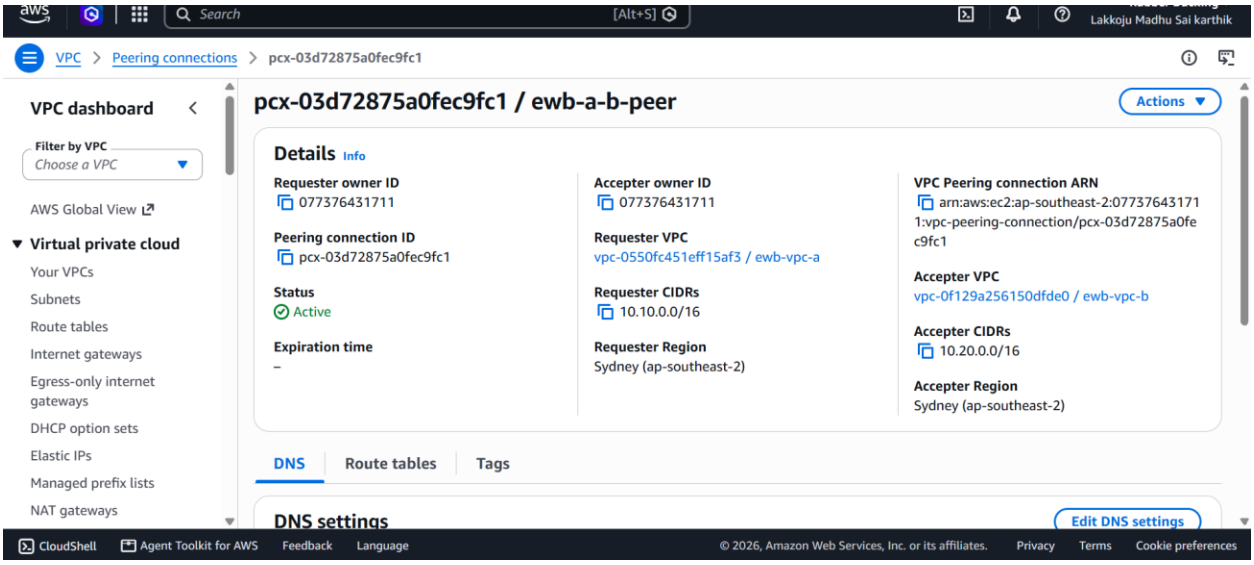
INSTANCE OF EWB-APP-A



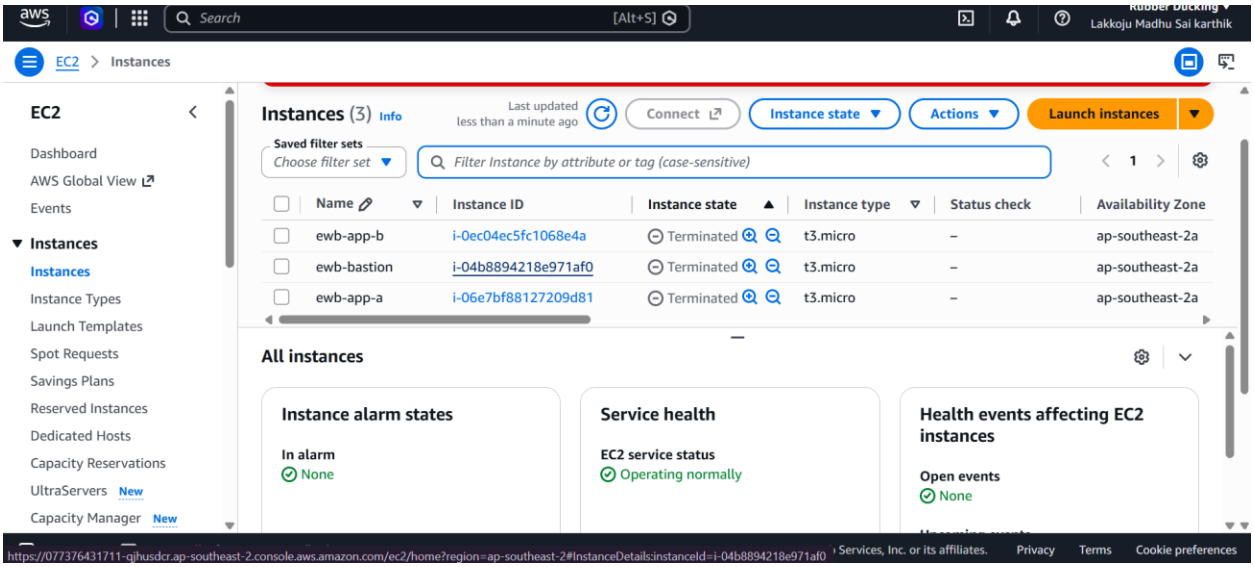
INSTANCE OF EWB-BASTION



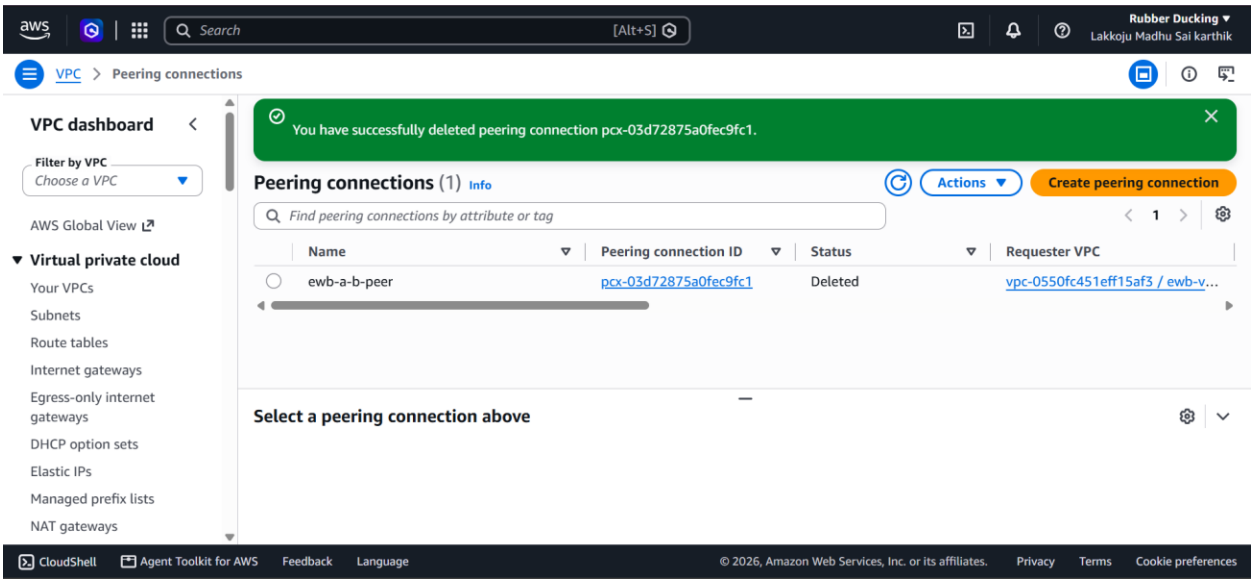
Internet gateway



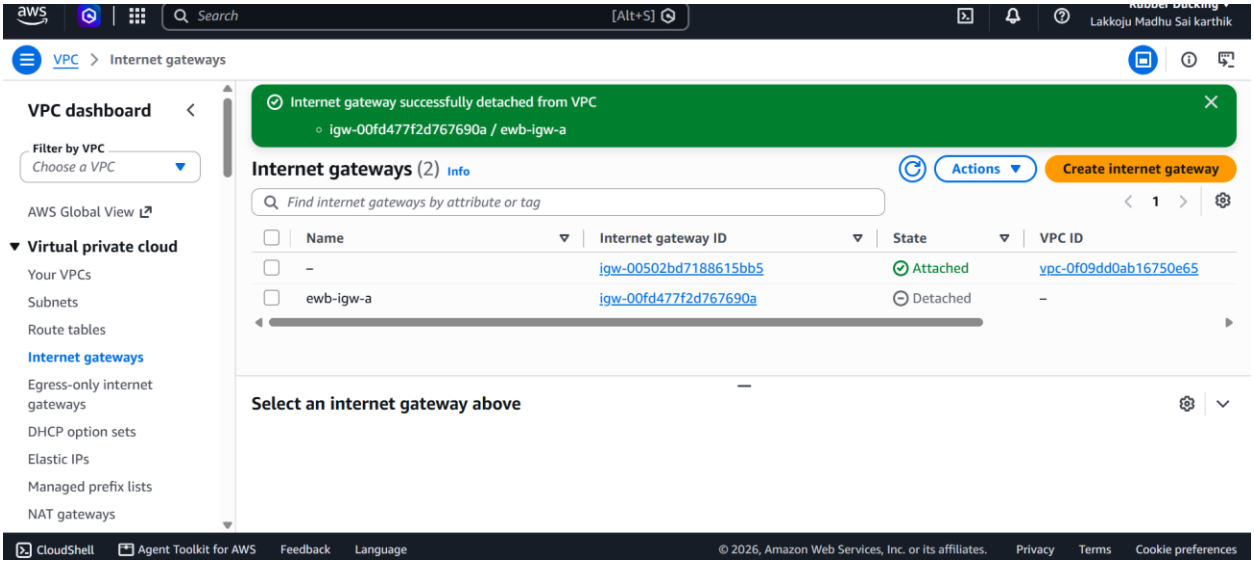
PEERING CONNECTIONS



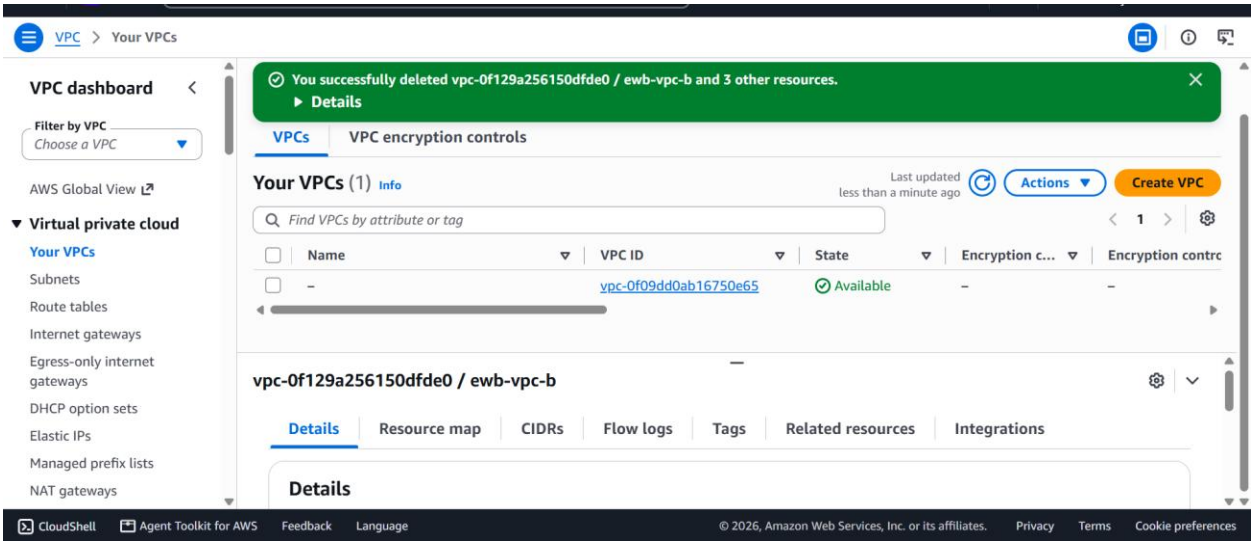
INSTANCES ARE TERMINATED



PEER CONNECTION ID DELETED



INTERNET GATEWAY IS DETACHED



AFTER VPC DELETED